Network Analysis from Scratch: Introduction to Gephi
Session 3 Handout

Brian Tsz Ho Wong, Ph.D.
Training Fellow, Edinburgh Futures Institute, University of Edinburgh
Email: twong4@ed.ac.uk

21 October 2026

Deciphering a Financial Network with Gephi

Today we will use Gephi to reconstruct and analyse the ownership networks in the contemporary Hong Kong stock market. On 17 May 2017, David Webb , a retired investment banker and activist investor, published an article on his website. The article, entitled 'The Enigma Network: 50 stocks not to own'[1]. Webb argued that investors should not invest in 50 HK-listed companies, which he said had cross-owned each other's shares. This allowed the 'big bosses' behind the 'Enigma Network' to easily manipulate the prices of these stocks, and ultimately damage investors' interests. Webb's article had shocked the market, and the Securities and Futures Commission began to investigate Webb's allegations. As a result, several executives of the companies in Webb's networks were arrested and prosecuted, although most were eventually acquitted.

Webb's website provided comprehensive data on the 'Enigma Network', especially the daily holdings data from the Central Clearing and Automated Settlement System (CCASS).[2] This tutorial utilises the CCASS data from 16 May 2017, the day before Webb's article was published, to reconstruct the 'Enigma Network' and to demonstrate Gephi's ability to analyse real-life financial data.

[1] https://web.archive.org/web/20250212225721/https://webb-site.com/articles/enigma.asp.
[2] https://webbsite.0xmd.com/dbpub/.

Cleaning CCASS Data in OpenRefine

1. Download ‘enigma_ccass_2017-05-16_details’ from GitHub.
2. Open OpenRefine, upload ‘enigma_ccass_2017-05-16_details’, and then click ‘Next’
3. Click on ‘Create projects’
4. All > Edit columns > Re-order/ remove columns
5. Click on ‘Remove all’, then drag ‘article_name’, ‘participant_name’, and ‘stack_pct’ under ‘Drag columns to re-order’.
6. Place ‘participant_name’ before ‘article_name’. Click ‘OK’.
7. All > Edit all columns > To titlecase. Only select the ‘article_name’, and ‘participant_name’ columns.
8. Click on the button next to ‘stack_pct’, select ‘Text facet’.
9. In the facet panel, click the value ‘0’ (4311 results),
10. All > Edit rows > Remove all matching rows.
11. In the facet panel, click ‘exclude’ next to ‘0’.
12. Export it as an .xlsx file. Rename it as ‘enigma_openrefine’.

Preparing a Gephi Dataset in Excel

1. Duplicate the ‘enigma_openrefine’ spreadsheet.
2. Copy all the entities in ‘participant_name’ to a new spreadsheet, then remove the duplicates.
3. Do the same for ‘article_name’.
4. Copy the filtered list of ‘article_name’ under that of ‘participant_name’, and then remove the duplicates
5. Add one row to the left and one to the top of the filtered entities. Assign an ID to each entity. Rename the spreadsheet ‘Nodes’ (nodes table).
6. Go back to the first spreadsheet. Add a column next to ‘participant_name’ and ‘article_name’, then copy the ‘nodes table’ to this spreadsheet.
7. Use the ‘xlookup’ function to match the entities to their IDs.

8. Open a new spreadsheet. Copy the IDs of the entities from the ‘participant_name’ to the first column. Rename this column ‘Source’.
9. Copy the IDs of the entities from ‘article_name’ to the second column. Rename this column ‘Target’. In this case, the entities in ‘participant_name’ are the shareholders of the companies in the ‘Enigma Network’. The entities in ‘article_name’ represent the 50 companies in the network.
10. Copy ‘stack_pct’ to the third column. Rename this column ‘Weight’. These numbers represent the percentage of shares held by the shareholders. A higher percentage would give an edge more weight.
11. Add a new column called ‘Type’ next to the ‘Weight’ column. Fill the cells in this column with ‘Directed’.
12. Copy the ‘Nodes’ and ‘Edges’ to a new Excel file and rename it ‘enigma_Gephi’.

Visualise and Analyse the ‘Enigma Network’ in Gephi

1. Download the ‘Enigma Network_Gephi.xlsx’ and open Gephi. Do not open the spreadsheets in Excel, if you have opened them, close them before opening Gephi.
2. Click on ‘New Project’ and then go to ‘Data Laboratory’.
3. Click on ‘Import Spreadsheet’, select ‘enigma_Gephi.xlsx’ and press ‘open’.
4. Import ‘Nodes’. Select ‘Directed’ for ‘Graph Type’.
5. Go to ‘Import spreadsheet’ and import ‘Edges’.

Check for any duplicate source-target pairs:

Prompt: Here is a Gephi dataset. Check for any duplicate source-target pairs. If so, identify and remove them from the dataset.

6. Click on ‘Import Spreadsheet’, select ‘enigma_Gephi_deduped.xlsx’ and press ‘open’.
7. Import ‘Nodes’. Select ‘Directed’ for ‘Graph Type’.
8. Go to ‘Import spreadsheet’ and import ‘Edges’.
9. Select ‘BigDecimal’ for the ‘Weight’ column, and then choose ‘Append to existing workplace’.
10. Go to ‘Overview’ and select ‘Force Atlas’ for the layout. Then reset the ‘Repulsion strength’ to '10000’.
11. Click ‘Run’.

Measuring the Network

1. Go to ‘Statistics’.
2. Run ‘Avg. Weighted Degree’
3. Go to ‘Data Laboratory’, and you will find statistics for several types of centrality measures, including ‘Weighted In-Degree’, ‘Weighted Out-Degree’, and ‘Weighted Degree’
4. We will focus on ‘Weighted Out-Degree’ centrality, as this represents the number of edges connected to each node, taking into account the weight of the edges. In our case, it measures a broker’s investment in or its shares of the 50 companies in the Enigma Network. In other words, the nodes with high degree in this measurement are probably the ‘big bosses’ behind the network.

Visualising the Network

Before we analyse the result of centrality measures, we can do some refinements to the network visualisation.

1. Go to ‘Overview’ and select ‘Filters’
2. Select ‘Edge Weight’ under ‘Edges’

3. Change the ‘Edge Weight’ range from ‘0.01 to 75.04’ to ‘1 to 75.04’.
4. Click on ‘Select’ and then on ‘Filter’
5. Click on the ‘Attributes’ button and select ‘company’. Then click on the bold ‘T’ next to the ‘Attributes’ button to make the names of the entities pop up.
6. Go to ‘Appearance’ and click on the ‘Size’ button
7. Select ‘Ranking’ and choose ‘Weighted Out-Degree’. Then, change the ‘Min size’ to ‘10’ and the ‘Max size’ to '80’
8. Click on the ‘Label Size’ button
9. Select ‘Ranking’ and choose ‘Weighted Out-Degree’. Then, change the ‘Min size’ to ‘0.5’ and the ‘Max size’ to '10’
10. Select ‘Label Adjust’ under ‘Layout’, then click ‘Run’
11. You can export the visualisation as a graph file, PDF or PNG file, or take a screenshot using the ‘Take screenshot’ button. Remember to adjust the width and height in the ‘Screenshot settings’ menu to improve the picture’s resolution.

Analysing the Network

1. Go to ‘Data Laboratory’
2. Re-order the ‘Weighted Out-Degree’ from largest to smallest
3. You can then identify the securities companies that masterminded the ‘Enigma Network’.

Interestingly, some of the nodes with a high weighted out-degree are major banks, such as HSBC and the Bank of China. These banks were certainly not involved in the conspiracy, other than holding shares in the 50 companies for various reasons. It is therefore useful to remove these banks from the dataset.

Refine the Dataset in OpenRefine

1. Upload the ‘enigma_openrefine’ spreadsheet to OpenRefine.
2. Select only the ‘enigma_openrefine’ spreadsheet.

3. Click on ‘Create projects’.
4. Click on the button next to ‘article_name’, select ‘Text facet’.
5. In the facet panel, click on the result ‘Bank of China (hong Kong) Ltd’.
6. All > Edit rows > Remove all matching rows.
7. In the facet panel, click ‘exclude’ next to ‘Bank of China (hong Kong) Ltd’.

You can do the same for other banks, such as Citibank and HSBC. Save the file as .xlsx and then repeat the above steps to prepare a Gephi dataset and network.